

Data Structures and Algorithms (No GenAI allowed) – Lab 2

Lab 2

- Create a test class for the **reference based** ADT list (Linked List) that tests all of the methods of this class (e.g. isEmpty, size, add, remove etc).
- Note for the two questions below (displayList and listLargest) you should **NOT** use the get() or find() method from ListReferenceBased, instead you should write your own loop to traverse the list as was shown in class (using a temporary Node reference like curr).
- Create a displayList() method **in the ListReferenceBased class** that displays or prints out the list in some simple way and call this method from the main method in the test class.
- Create a listLongest() method **in the ListReferenceBased class** that finds the longest String object in the list and returns it. Call this method from the main method in the test class and print the result. By longest we mean the string with the most characters. You can assume the list contains String objects.

As you complete each part you should commit changes to your Git repo. Of course you can also make commits even if you partially complete a section (e.g. you have worked on the lab, made some progress but not quite completed the section – still commit your work).

Upload your Java files and your git_history.txt file to Brightspace. You can generate your git_history.txt file using the git log command from a Terminal inside your repo root directory, e.g. **git log > git_history.txt**

If you are using eclipse make sure the terminal plugin is installed. (Keyboard shortcut CTRL-ALT-T should open the terminal). If not then install it via the Eclipse marketplace in the menu bar (Help->Eclipse Marketplace and search for Terminal). The approach should be similar for other IDE's.